Geometric Pearson Assignment Bounds
Extended Derivation and Comparison

1 Classification Model and Notation

1.1 Fixed-bank classification model

Let $P_1, \dots, P_R$ be distinct fixed regime distributions. Reference i contains $X_{i,1}, \dots, X_{i,n_i}$, and the query contains $Y_1, \dots, Y_q$ from a fixed distribution Q. Sampling assumptions are stated in Assumption 6.1.

Let $\mathcal{H}_k$ be a real, separable RKHS with measurable feature map $\phi(x) = k(\cdot, x)$. Assume that k is positive semidefinite and

$$\sup_x k(x,x) \le K, \qquad 0 < K < \infty. \tag{1}$$

Thus $\|\phi(x)\| \le \sqrt{K}$. The normalized Gaussian kernel has $K = 1$. Pointwise nonnegativity $k(x,y) \ge 0$ is *not* required for the main deterministic or Markov arguments. Whenever it is used to improve a constant, we say so explicitly.

For any distribution P, define its population moments

$$\mu_P = \mathbb{E}_P \phi(X), \qquad T_P = \mathbb{E}_P[\phi(X) \otimes \phi(X)], \tag{2}$$

where $(u \otimes v)w = \langle v, w \rangle u$. Boundedness and measurability ensure the relevant expectations exist. For reference i define

$$\begin{aligned} \widehat{\mu}_i &= \frac{1}{n_i} \sum_{a=1}^{n_i} \phi(X_{i,a}), & \widehat{T}_i &= \frac{1}{n_i} \sum_{a=1}^{n_i} \phi(X_{i,a}) \otimes \phi(X_{i,a}), \\ \widehat{\mu}_Q &= \frac{1}{q} \sum_{b=1}^{q} \phi(Y_b), & \Delta_i &= \mu_Q - \mu_{P_i}, \qquad \widehat{\Delta}_i = \widehat{\mu}_Q - \widehat{\mu}_i. \end{aligned} \tag{3}$$

T_P is uncentered; subtracting $\mu_P \otimes \mu_P$ gives a different covariance operator and a different score.

For fixed $\lambda > 0$, put

$$\begin{aligned} A_i &= T_{P_i} + \lambda I, & \widehat{A}_i &= \widehat{T}_i + \lambda I, \\ D_i &= D_\lambda(Q; P_i) = \tfrac{1}{2} \left\langle \Delta_i, A_i^{-1} \Delta_i \right\rangle, & \widehat{D}_i &= \tfrac{1}{2} \left\langle \widehat{\Delta}_i, \widehat{A}_i^{-1} \widehat{\Delta}_i \right\rangle. \end{aligned} \tag{4}$$

The reference determines the inverse operator, so the score is directed. For a positive operator B, write $\|v\|_B^2 = \langle v, Bv \rangle$.

The classifier uses a common threshold $\tau \ge 0$:

$$i^\star_{\text{emp}} \in \operatorname*{arg\,min}_{1 \le i \le R} \widehat{D}_i, \qquad \begin{cases} \text{recall } i^\star_{\text{emp}}, & \min_i \widehat{D}_i \le \tau, \\ \text{create a new identity}, & \min_i \widehat{D}_i > \tau. \end{cases} \tag{5}$$

Successful classification below gives a unique accepted reference for a known regime, so tie breaking is irrelevant on the success event.

1.2 Three different notions of separation

We keep the following quantities distinct:

$$d_i = \|\mu_Q - \mu_{P_i}\|, \qquad s_i = \sqrt{2D_\lambda(Q;P_i)}, \qquad D_i = s_i^2/2. \tag{6}$$

d_i is an ordinary RKHS distance, s_i is a reference-weighted distance to the query, and D_i is the discrepancy itself. A margin denoted by δ in one analysis must not be substituted into another without checking which quantity it bounds. Coherence in the ICASSP manuscript uses distances between *normalized directions*; those are different again.

2 Universal operator bounds and the Pearson–MMD connection

Lemma 2.1 (Second-moment order)**.** *For every population or empirical reference,*

$$0 \preceq T_{P_i}, \widehat{T}_i \preceq KI, \qquad \lambda I \preceq A_i, \widehat{A}_i \preceq (K+\lambda)I. \tag{7}$$

In particular, all regularized operators are boundedly invertible and

$$\frac{1}{K+\lambda} I \preceq A_i^{-1}, \widehat{A}_i^{-1} \preceq \frac{1}{\lambda} I. \tag{8}$$

Proof. For $f \in \mathcal{H}_k$, the reproducing property and Cauchy–Schwarz give

$$0 \le \langle f, T_{P_i} f\rangle = \mathbb{E}_{P_i}|f(X)|^2 \le \mathbb{E}_{P_i} \|f\|^2 \|\phi(X)\|^2 \le K \|f\|^2 .$$

Replace the expectation by a finite average for the empirical operator. Adding λI proves (7). Positive bounded operators with lower bound λI have bounded inverses; inversion reverses operator order, proving (8). □

Corollary 2.2 (Empirical and population score sandwiches)**.** *For every realized data set,*

$$\boxed{\frac{\left\|\widehat{\Delta}_i\right\|^2}{2(K+\lambda)} \le \widehat{D}_i \le \frac{\left\|\widehat{\Delta}_i\right\|^2}{2\lambda}.} \tag{9}$$

The same inequalities hold for D_i and Δ_i. In particular,

$$\frac{d_i}{\sqrt{K+\lambda}} \le s_i \le \frac{d_i}{\sqrt{\lambda}}. \tag{10}$$

Proof. Apply (8) to the corresponding mean difference and divide by two. Taking nonnegative square roots gives the last statement. □

These bounds hold for the realized empirical vector even though $\widehat{\Delta}_i$ and $\widehat{A}_i$ use the same reference observations. No independence between these estimates is assumed or needed.

Remark 2.3 (What has been removed, and what has not). The empirical second moment is still present in the implemented score. The analysis bounds its inverse using universal spectral endpoints rather than estimating $\left\|\widehat{T}_i - T_{P_i}\right\|_{\mathrm{op}}$. This does not replace the algorithm's second moment by the identity.

For a strictly positive boundedly invertible B, define $D_B = \langle \Delta, B^{-1}\Delta\rangle/2$. Then

$$2D_I = \|\Delta\|^2 = \mathrm{MMD}^2(P,Q). \tag{11}$$

Thus MMD is the identity-metric specialization of the weighted family. The literal substitution $T_P = I$ into the regularized Pearson score instead gives $D_\lambda = \mathrm{MMD}^2/[2(1+\lambda)]$. Under a bounded

kernel, $\operatorname{tr}(T_P) = \mathbb{E}k(X, X) \le K$, so $T_P = I$ is generally unavailable in an infinite-dimensional RKHS. It is a formal specialization, not a sampling assumption required by the analysis.

For fixed P, Q, the sandwich also gives

$$\frac{\lambda}{K+\lambda} \|\Delta\|^2 \le 2\lambda D_\lambda(Q; P) \le \|\Delta\|^2 . \tag{12}$$

Consequently $2\lambda D_\lambda \to \mathrm{MMD}^2$ as $\lambda \to \infty$. The identical statement holds empirically for fixed batches.

3 Deterministic embedding-error bounds

3.1 Individual and mean-difference error events

Suppose that, for specified nonnegative radii,

$$\mathcal{E} = \{\|\widehat{\mu}_Q - \mu_Q\| \le \varepsilon_Q,\ \|\widehat{\mu}_i - \mu_{P_i}\| \le \varepsilon_i \text{ for every } i\} . \tag{13}$$

On this event,

$$\left\|\widehat{\Delta}_i - \Delta_i\right\| \le \varepsilon_Q + \varepsilon_i =: r_i. \tag{14}$$

More generally, the following results only need the error bounds $\left\|\widehat{\Delta}_i - \Delta_i\right\| \le r_i$; separate bounds on the individual embeddings are one way to obtain them. This distinction will give a sharper Markov calibration later.

Lemma 3.1 (Empirical distance error)**.** *If* $\left\|\widehat{\Delta}_i - \Delta_i\right\| \le r_i$*, then*

$$[d_i - r_i]_+ \le \left\|\widehat{\Delta}_i\right\| \le d_i + r_i. \tag{15}$$

Proof. Write $\widehat{\Delta}_i = \Delta_i + (\widehat{\Delta}_i - \Delta_i)$ and use the triangle inequality for the upper bound. The reverse triangle inequality gives $\left\|\widehat{\Delta}_i\right\| \ge d_i - r_i$. Norms are nonnegative, so the lower bound may be replaced by its positive part. □

Theorem 3.2 (Geometric Pearson bounds)**.** *For every reference satisfying* $\left\|\widehat{\Delta}_i - \Delta_i\right\| \le r_i$*,*

$$\boxed{\frac{[d_i - r_i]_+^2}{2(K+\lambda)} \le \widehat{D}_i \le \frac{(d_i + r_i)^2}{2\lambda}.} \tag{16}$$

Proof. All terms in (15) are nonnegative. Square its inequalities and substitute into (9). □

The proof contains only spectral order and triangle inequalities. It is valid for unequal window lengths and for dependent, selected, or even deterministic batches *provided the displayed error bounds hold*. Those cases do not automatically have the i.i.d. probabilities derived later.

3.2 Equal individual radii

If every individual embedding error is at most ε, take $r_i = 2\varepsilon$. Theorem 3.2 becomes

$$\boxed{\frac{[d_i - 2\varepsilon]_+^2}{2(K+\lambda)} \le \widehat{D}_i \le \frac{(d_i + 2\varepsilon)^2}{2\lambda}.} \tag{17}$$

The factor two comes from estimating both the query mean and the stored mean. It would be inappropriate to remove that factor while still using the same individual error radius.

For a known regime $Q = P_j$, $d_j = 0$, and hence

$$\widehat{D}_j \le \frac{2\varepsilon^2}{\lambda}. \tag{18}$$

For a wrong reference with $d_i \ge \delta$,

$$\widehat{D}_i \ge \frac{[\delta - 2\varepsilon]_+^2}{2(K+\lambda)}. \tag{19}$$

For a normalized Gaussian kernel, take $K = 1$.

3.3 Why a single global whitening argument is invalid

For each fixed i, $\widehat{D}_i$ is half a squared norm induced by $\widehat{A}_i^{-1}$. However, different references use different operators. There is generally no single whitening map that turns all comparisons into ordinary distances in a common transformed space. Nor is a directed Pearson score generally a symmetric metric on distributions. The above argument is legitimate because each comparison is bounded separately, using spectral inequalities valid for its own operator.

The original coherence proof normalizes nonzero population embeddings and therefore assumes a lower bound on their norms. No corresponding nondegeneracy assumption is needed here: we do not divide by a mean norm. The positive λ stabilizes the inverse operator instead.

4 Correct assignment and novelty rejection

Assumption 4.1 (Mean-embedding separation)**.** Let $\mathcal{Q}$ be an admissible family of query distributions. There is $\delta_* > 0$ such that, for each fixed $Q \in \mathcal{Q}$,

$$\|\mu_Q - \mu_{P_i}\| \ge \delta_* \quad \text{whenever } Q \ne P_i. \tag{20}$$

For a single known query $Q = P_j$, it is enough to use $\delta_j = \min_{i \ne j} \left\|\mu_{P_j} - \mu_{P_i}\right\|$. For a new query, use $\delta_{\mathrm{new}}(Q) = \min_i \|\mu_Q - \mu_{P_i}\|$. A margin within the stored bank alone does not control an arbitrary new distribution. If the bank has one regime and the query is known, there are no wrong references to reject; the wrong-reference condition is then vacuous.

Assumption 4.1 is an explicit restriction on the admissible comparisons. A characteristic kernel distinguishes different distributions, but does not supply a positive uniform separation over all possible distinct distributions. The probability claims below are for each fixed admissible Q, not simultaneous statements over all possible queries.

Theorem 4.2 (Correct assignment on a simultaneous accuracy event)**.** *Suppose* $\left\|\widehat{\Delta}_i - \Delta_i\right\| \le r$ *for every reference. For a known query* $Q = P_j$*, the following statements hold.*

1. *If* $r \le \sqrt{2\lambda\tau}$*, the correct reference is accepted; in particular, the algorithm does not declare a new regime.*

2. *If additionally*

$$\delta_j > r + \sqrt{2(K+\lambda)\tau}, \tag{21}$$

every wrong reference is rejected. The accepted set is exactly $\{j\}$ *and the minimum-score rule recalls* j *uniquely.*

For a new query, the condition

$$\delta_{\mathrm{new}}(Q) > r + \sqrt{2(K+\lambda)\tau} \tag{22}$$

ensures that every reference is rejected and a new identity is created.

Proof. For $Q = P_j$, the population mean difference to j vanishes, so Theorem 3.2 gives $\widehat{D}_j \leq r^2/(2\lambda)$. The first condition makes this at most τ. For $i \neq j$, (21) implies $d_i - r > \sqrt{2(K+\lambda)\tau}$. In particular $d_i - r > 0$, and (16) gives $\widehat{D}_i > \tau$. Thus the correct reference passes and all others fail. For a new query, apply the same lower-bound argument to every reference using (22). □

Remark 4.3 (Acceptance versus identity). Acceptance of the correct reference alone prevents false novelty, but does not exclude a smaller score at an incorrect reference. The second condition is needed for the correct identity. Conversely, rejection of all wrong references alone does not ensure acceptance of the correct one.

5 Threshold interval and its exact algebra

For a common error radius r and assumed margin $\delta > 0$, define

$$U(r) = \frac{r^2}{2\lambda}, \qquad L(r,\delta) = \frac{[\delta - r]_+^2}{2(K+\lambda)}. \tag{23}$$

A sufficient common threshold interval is

$$\boxed{U(r) \leq \tau < L(r,\delta).} \tag{24}$$

The lower endpoint is included because acceptance uses $\leq \tau$. The upper endpoint is excluded because equality in the lower score bound would not ensure rejection.

Proposition 5.1 (Nonemptiness of the geometric certificate)**.** *If $r > 0$, interval* (24) *is nonempty if and only if*

$$\boxed{\delta > r\left(1 + \sqrt{\frac{K+\lambda}{\lambda}}\right).} \tag{25}$$

For $r = 0$, it is nonempty for every $\delta > 0$.

Proof. For $r > 0$, $L > U$ is equivalent to $[\delta - r]_+ > r\sqrt{(K+\lambda)/\lambda}$. The right side is positive, so this forces $\delta - r > 0$ and the positive part may be removed. Rearrange to obtain (25). All steps are reversible. If $r = 0$, $U = 0$ and $L = \delta^2/[2(K+\lambda)] > 0$. □

With individual radius ε, substitute $r = 2\varepsilon$:

$$\frac{2\varepsilon^2}{\lambda} \leq \tau < \frac{(\delta - 2\varepsilon)^2}{2(K+\lambda)}, \tag{26}$$

$$\delta > 2\varepsilon\left(1 + \sqrt{\frac{K+\lambda}{\lambda}}\right). \tag{27}$$

The interval formula presupposes the last inequality, which ensures that the squared term is not being used on a negative argument.

Remark 5.2 (Necessary for the certificate, not for the classifier). Nonemptiness is equivalent to (25) for these particular upper and lower bounds. It is not a necessary condition for successful classification: the realized scores can separate even when these worst-case bounds overlap.

5.1 A regularization condition with a fixed mean margin

For $r > 0$, condition (25) can hold for some finite λ only if $\delta > 2r$. If that is true, it is equivalent to

$$\boxed{\lambda > \frac{Kr^2}{\delta(\delta - 2r)}.} \tag{28}$$

Indeed, divide (25) by r, subtract one, and square the positive sides to obtain $K/\lambda < \delta(\delta-2r)/r^2$. This explains why very small regularization makes this uniform certificate demanding. It does not prove that increasing λ improves the actual classifier.

Writing $a = 2\lambda\tau$, the interval becomes

$$r^2 \le a < \frac{\lambda}{K+\lambda}(\delta - r)^2. \tag{29}$$

As $\lambda \to \infty$ it approaches the ordinary squared-embedding distance interval $r^2 \le a < (\delta - r)^2$, consistent with the MMD limit (12). Its nonemptiness limit is $\delta > 2r$.

6 Probability guarantees without McDiarmid

6.1 The exact embedding variance calculation

Assumption 6.1 (Within-batch sampling)**.** Each reference batch consists of i.i.d. observations from its fixed P_i, and the query consists of i.i.d. observations from a fixed Q. The kernel and λ are fixed independently of the evaluated batches. For the individual-embedding calibration below, independence across batches is not needed. For the direct mean-difference calibration we additionally require the query batch to be independent of each reference batch.

Lemma 6.2 (Mean-square embedding error)**.** *For N i.i.d. samples from a distribution P,*

$$\mathbb{E}\left\|\widehat{\mu}_P - \mu_P\right\|^2 = \frac{\mathbb{E}_P k(X,X) - \|\mu_P\|^2}{N} \le \frac{K}{N}. \tag{30}$$

Proof. Let $V_a = \phi(X_a) - \mu_P$. Then $\mathbb{E}V_a = 0$ and

$$\mathbb{E}\left\|N^{-1}\sum_a V_a\right\|^2 = N^{-2}\sum_{a,b}\mathbb{E}\left\langle V_a, V_b\right\rangle.$$

For $a \ne b$, independence and centering give zero cross terms. The N diagonal terms equal $\mathbb{E}\left\|\phi(X) - \mu_P\right\|^2$. Expanding that norm gives $\mathbb{E}k(X,X) - \|\mu_P\|^2$, proving the identity and bound. □

Centering here is an algebraic identity; it does not assume $\mu_P = 0$. Nor does (30) assert a pointwise bound $\left\|\phi(X) - \mu_P\right\|^2 \le K$.

6.2 The individual-embedding union bound

Apply Markov's inequality to the nonnegative variable $\left\|\widehat{\mu}_P - \mu_P\right\|^2$:

$$\mathbb{P}\{\left\|\widehat{\mu}_P - \mu_P\right\| > \varepsilon\} \le \frac{K}{N\varepsilon^2}, \qquad \varepsilon > 0. \tag{31}$$

With $n_i = q = n$, a union bound over $R+1$ embeddings gives

$$\mathbb{P}(\mathcal{E}_\varepsilon^c) \le \frac{K(R+1)}{n\varepsilon^2}, \tag{32}$$

where $\mathcal{E}_\varepsilon$ is (13) with all radii equal. No independence between the error events is required.

Corollary 6.3 (Markov-only fixed-bank certificate)**.** *For $\rho \in (0,1)$, set*

$$\varepsilon_n = \sqrt{\frac{K(R+1)}{n\rho}}, \qquad r_n = 2\varepsilon_n. \tag{33}$$

For each fixed admissible query, $\mathcal{E}_{\varepsilon_n}$ holds with probability at least $1-\rho$. Any deterministic threshold satisfying $U(r_n) \le \tau < L(r_n, \delta_*)$ gives correct known-regime recall or correct novel-regime rejection with that probability, as appropriate to the query. A sufficient batch size for a nonempty interval is

$$\boxed{n > \frac{4K(R+1)}{\rho\delta_*^2}\left(1+\sqrt{\frac{K+\lambda}{\lambda}}\right)^2.} \tag{34}$$

Proof. Substitute (33) into (32) and apply Theorem 4.2. Finally square (25) with $r = r_n$. All sides are nonnegative, yielding (34). □

The probability is joint over the sampled memory and query. It is not a guarantee conditional on every possible realized memory. The factor $R+1$ comes from the chosen union bound, not an established intrinsic complexity of recurrent-regime learning.

6.3 Unequal batch sizes and confidence allocation

For n_i and q not necessarily equal, the common individual radius

$$\varepsilon = \sqrt{\frac{K}{\rho}\left(\frac{1}{q} + \sum_{i=1}^{R}\frac{1}{n_i}\right)} \tag{35}$$

still makes the simultaneous event hold with probability at least $1-\rho$, by summing (31). Alternatively choose $\alpha_Q, \alpha_1, \ldots, \alpha_R > 0$ with sum at most ρ, and set

$$\varepsilon_Q = \sqrt{K/(q\alpha_Q)}, \qquad \varepsilon_i = \sqrt{K/(n_i\alpha_i)}.$$

Then use the reference-specific $r_i = \varepsilon_Q + \varepsilon_i$ in Theorem 3.2. A common threshold can be certified using the largest necessary null upper bound and the smallest wrong-reference lower bound. The allocation should be specified independently of the realized scores when claiming a static threshold.

7 A sharper Markov route using mean differences directly

The previous calibration first bounds two mean errors and adds their norms. That is convenient, but not the only option. Under the cross-batch independence in Assumption 6.1, the two centered mean errors have zero expected cross inner product. Therefore

$$\begin{aligned}\mathbb{E}\left\|\widehat{\Delta}_i - \Delta_i\right\|^2 &= \mathbb{E}\left\|\widehat{\mu}_Q - \mu_Q\right\|^2 + \mathbb{E}\left\|\widehat{\mu}_i - \mu_{P_i}\right\|^2 \\ &\le K\left(\frac{1}{q} + \frac{1}{n_i}\right).\end{aligned} \tag{36}$$

This does not assert independence between the mean difference and its reference operator; they may remain coupled.

Proposition 7.1 (Direct simultaneous mean-difference radius)**.** *Let*

$$b = \sqrt{\frac{K}{\rho}\left(\frac{R}{q} + \sum_{i=1}^{R}\frac{1}{n_i}\right)}. \tag{37}$$

With probability at least $1-\rho$, $\left\|\widehat{\Delta}_i - \Delta_i\right\| \le b$ for every i.

Proof. Markov's inequality and (36) bound failure for pair i by $K(1/q + 1/n_i)/b^2$. Sum over i. Scores share the query, but independence between pairwise failure events is not needed. □

For equal sizes,

$$b_n = \sqrt{\frac{2KR}{n\rho}}, \qquad \boxed{\frac{b_n^2}{2\lambda} \le \tau < \frac{(\delta_* - b_n)^2}{2(K+\lambda)}} \tag{38}$$

is a sufficient threshold interval whenever it is nonempty. The sufficient sample size becomes

$$\boxed{n > \frac{2KR}{\rho\delta_*^2}\left(1 + \sqrt{\frac{K+\lambda}{\lambda}}\right)^2.} \tag{39}$$

This replaces $4(R+1)$ in (34) by $2R$. The gain comes from bounding the mean difference directly, not from changing the score, the geometric proof, or the probability inequality.

7.1 Known query versus unknown query

The same simultaneous event controls all R comparisons. For a known query, it ensures one correct acceptance and $R-1$ wrong-reference rejections. For an unknown query, it ensures R rejections. One should not add the known-query and unknown-query failure probabilities as if they were two events under the same query distribution: they describe different fixed-query scenarios.

7.2 An analytic numerical illustration

Take $K = 1$, $R = 4$, $\rho = 0.05$, $\delta_* = 0.5$, $\lambda = 0.1$ and $n = 50{,}000$. These are hypothetical parameters for evaluating the formulas, not measured regime separations or experimental outcomes. Individual-embedding Markov calibration gives

$$\varepsilon_n \simeq 0.0447214, \qquad r_n \simeq 0.0894427, \qquad U \simeq 0.0400000, \quad L \simeq 0.0766169.$$

Thus $\tau = 0.0583$ lies inside the certified interval. The direct mean-difference route gives $b_n \simeq 0.0565685$, and the wider interval

$$0.0160000 \le \tau < 0.0893779.$$

The respective sufficient sample-size cutoffs are $n > 29{,}813.20$ and $n > 11{,}925.28$. These conservative sufficient cutoffs are not requirements for the algorithm to work in practice.

7.3 Optional comparison with the original concentration step

McDiarmid is unnecessary for the deterministic theorem, but can improve the confidence dependence. For a general bounded kernel, the elementary bounded-differences calculation in Appendix B gives the single-embedding radius

$$\varepsilon_{\mathrm{BD}}(n, a) = \sqrt{K/n}\left(1 + \sqrt{2\log(1/a)}\right). \tag{40}$$

Choosing $a = \rho/(R+1)$ and using a union bound permits the exact same geometric theorem with $r = 2\varepsilon_{\mathrm{BD}}$. The corresponding sufficient sample size is

$$n > \frac{4K}{\delta_*^2}\left(1 + \sqrt{2\log\frac{R+1}{\rho}}\right)^2\left(1 + \sqrt{\frac{K+\lambda}{\lambda}}\right)^2. \tag{41}$$

For the hypothetical parameters above this cutoff is approximately 4,853.60. The improvement relative to Markov reflects exponential rather than second-moment tails; it does not alter the geometric argument. The source ICASSP radius has an additional factor two inside its logarithm and is also a valid, slightly more conservative choice.

8 Keeping the reference geometry: operator-aware bounds

The simple analysis uses only λ and $K+\lambda$ as spectral endpoints. To compare directly to the population Pearson discrepancy, consider one pair and suppress the reference subscript. Assume the event

$$\left\|\widehat{\Delta}-\Delta\right\| \le b, \qquad \left\|\widehat{T}-T\right\|_{\mathrm{op}} \le e. \tag{42}$$

Here $b, e \ge 0$ may be deterministic error radii. Define $c = 1 + e/\lambda$.

Lemma 8.1 (Relative inverse comparison)**.** *On* (42)*,*

$$c^{-1}A^{-1} \preceq \widehat{A}^{-1} \preceq cA^{-1}. \tag{43}$$

Proof. Self-adjointness gives $-eI \preceq \widehat{T}-T \preceq eI$. Since $A \succeq \lambda I$, $\widehat{A} \preceq A + eI \preceq cA$. Conversely, $A \preceq \widehat{A} + eI \preceq c\widehat{A}$ because $\widehat{A} \succeq \lambda I$. Invert both positive-operator inequalities to obtain (43). No commutativity is required. □

Theorem 8.2 (Population-Pearson comparison)**.** *Let* $s = \sqrt{2D_\lambda(Q;P)} = \|\Delta\|_{A^{-1}}$*. On event* (42)*,*

$$\boxed{\frac{\left[s - b/\sqrt{\lambda}\right]_+}{\sqrt{c}} \le \sqrt{2\widehat{D}_\lambda} \le \sqrt{c}\,(s + b/\sqrt{\lambda}).} \tag{44}$$

Under $Q = P$*, the sharper null upper bound is* $\widehat{D}_\lambda \le b^2/(2\lambda)$*.*

Proof. By Lemma 8.1,

$$c^{-1/2}\left\|\widehat{\Delta}\right\|_{A^{-1}} \le \left\|\widehat{\Delta}\right\|_{\widehat{A}^{-1}} \le c^{1/2}\left\|\widehat{\Delta}\right\|_{A^{-1}}.$$

The reverse and forward triangle inequalities in the fixed norm induced by A^{-1} give

$$\left[s - \left\|\widehat{\Delta}-\Delta\right\|_{A^{-1}}\right]_+ \le \left\|\widehat{\Delta}\right\|_{A^{-1}} \le s + \left\|\widehat{\Delta}-\Delta\right\|_{A^{-1}}.$$

Because $A^{-1} \preceq \lambda^{-1}I$, the error term is at most $b/\sqrt{\lambda}$. Combine these displays. If $Q = P$, use $\Delta = 0$ and the empirical inverse bound directly; this avoids the extra factor c in the upper bound. □

Suppose every incorrect comparison has $s_i \ge s_* > 0$, and event (42) holds for all references with common b, e. The sufficient recognition interval is

$$\boxed{\frac{b^2}{2\lambda} \le \tau < \frac{\left[s_* - b/\sqrt{\lambda}\right]_+^2}{2(1+e/\lambda)}.} \tag{45}$$

For $b > 0$, it is nonempty exactly when

$$s_* > \frac{b}{\sqrt{\lambda}}\left(1 + \sqrt{1+e/\lambda}\right). \tag{46}$$

If using a discrepancy margin $D_i \ge \eta_* > 0$ instead, substitute $s_* = \sqrt{2\eta_*}$. This reconciles the notation with the operator-aware interval in the existing extended notes. For the earlier conversation's individual mean radii, substitute $b = 2\varepsilon$.

8.1 Operator control without McDiarmid

Let $V(X) = \phi(X) \otimes \phi(X)$. The rank-one identities are

$$\|V(X)\|_{\mathrm{HS}} = k(X, X) \le K, \qquad \langle V(x), V(y)\rangle_{\mathrm{HS}} = k(x, y)^2. \tag{47}$$

The same Hilbert-space variance calculation as before gives

$$\mathbb{E}\left\|\widehat{T} - T\right\|_{\mathrm{HS}}^2 = \frac{\mathbb{E}k(X, X)^2 - \|T\|_{\mathrm{HS}}^2}{n} \le \frac{K^2}{n}. \tag{48}$$

Since the operator norm is at most the Hilbert–Schmidt norm, Markov gives

$$\mathbb{P}\{\left\|\widehat{T} - T\right\|_{\mathrm{op}} > e\} \le \frac{K^2}{ne^2}. \tag{49}$$

For independent equal-size reference and query batches, choose $\alpha_\mu, \alpha_T > 0$ and

$$b = \sqrt{\frac{2K}{n\alpha_\mu}}, \qquad e = \frac{K}{\sqrt{n\alpha_T}}. \tag{50}$$

Then (42) holds with probability at least $1 - \alpha_\mu - \alpha_T$. The two estimates share reference samples; the union bound does not require their independence. For a fixed bank, allocate the total failure probability across the relevant mean and operator events. This route remains McDiarmid-free, but it requires operator estimation control that the geometric route avoids.

8.2 Why operator-aware does not mean uniformly tighter

For the same mean radius b, the two lower bounds have different forms:

$$\begin{aligned} L_{\mathrm{geom}} &= \frac{[d - b]_+^2}{2(K + \lambda)}, \\ L_{\mathrm{op}} &= \frac{\left[s - b/\sqrt{\lambda}\right]_+^{,2}}{2(1 + e/\lambda)}. \end{aligned} \tag{51}$$

The second incorporates the actual population geometry, but pays an operator-estimation factor and a weighted mean-error term. Neither bound is asserted to dominate at every finite sample size. On a joint event where both apply, their maximum is also a valid lower bound.

For fixed P, Q, λ, as $b, e \to 0$,

$$L_{\mathrm{geom}} \longrightarrow \frac{d^2}{2(K + \lambda)} \le D_\lambda(Q; P), \qquad L_{\mathrm{op}} \longrightarrow D_\lambda(Q; P). \tag{52}$$

Thus the geometric analysis can retain a spectral slack even with perfectly estimated means. This is the precise cost of replacing the reference spectrum by the universal endpoints.

9 Interpretation, conservatism, and what is actually certified

9.1 Sources of looseness

The geometric certificate can lose accuracy at several separate steps:

1. Replacing the inverse operator by I/λ or $I/(K + \lambda)$ ignores the orientation of the actual mean difference.

2. Bounding two individual mean errors and adding their norms loses possible cancellation; direct mean-difference control partly avoids this.

3. Markov uses only a second moment and has weak tail dependence on confidence. Replacing it with an exponential inequality can reduce the required sample-size certificate without changing the geometry.

4. A union bound over a bank can overestimate the probability of at least one failure, especially when comparisons share the query.

5. A common margin and common threshold use the worst reference.

These are sufficient-error guarantees, not accurate predictions of the algorithm's empirical failure probability.

9.2 What the bound does not establish about Pearson versus MMD

Both classifiers compare differences of kernel mean embeddings. For a nonlinear characteristic kernel, equal raw means and variances do not imply equal kernel embeddings. The geometric certificate therefore does not identify MMD as intrinsically blind to higher-order distributional changes. Nor does it prove Pearson has better power or recognition accuracy. It proves that the actual Pearson classifier succeeds under an explicit margin and simultaneous empirical accuracy event. Whether reference weighting helps at matched false-alarm and false-merge operating points requires empirical evidence or a more specific model analysis.

The lower-bound regularization condition (28) also concerns this certificate. It is not an optimal statistical tuning rule. In an operator-aware analysis, s_i itself depends on λ; one must not vary λ while treating the population Pearson margin as an unrelated fixed constant.

9.3 A known memory versus an arbitrary realized memory

In the probability results, the memory is random under a declared sampling model even though its size and source distributions are fixed. The error probability is averaged jointly over that memory and the query. Conditioning on a very inaccurate realized memory can change the guarantee. Similarly, an algorithmically selected or replaced representative cannot be assumed to have the law of an unconditional i.i.d. reference batch without proof.

9.4 Temporal dependence, stopping times, and repeated comparisons

With dependent observations, the second-moment calculation contains additional terms:

$$\mathbb{E}\,\|\widehat{\mu}-\mu\|^2=\frac{1}{n^2}\sum_{a,b=1}^{n}\mathbb{E}\,\langle\phi(X_a)-\mu,\phi(X_b)-\mu\rangle\,. \tag{53}$$

The off-diagonal terms need not vanish. Using the i.i.d. radius, or simply substituting an asserted effective sample size, is unjustified without control of these terms or an appropriate dependence theorem.

Fresh samples after a stopping time can retain useful independence under suitable within-regime assumptions, but purity, regime duration, and the law of stored selected batches still require analysis. Requiring several consecutive threshold crossings does not create independent tests. The present results are per query and fixed bank; they do not give an unlimited-horizon sequential false-alarm probability. For a finite list of valid comparisons one may allocate failure probabilities and apply a union bound, but this is useful only after verifying each comparison's sampling assumptions.

The deterministic theorem remains available in all these situations if one can supply valid simultaneous error radii by another argument.

9.5 Kernel selection and preprocessing

A fixed kernel may be chosen using independent calibration data and then conditioned on for analysis, provided the assumptions hold under that conditioning. Fitting a bandwidth or PCA representation on the same reference batch does not automatically preserve the fixed-kernel i.i.d. concentration calculation. Historical kernels that differ between regimes also change the common-distance separation used by the simplest theorem. These issues are distinct from the algebraic validity of the empirical score and its universal spectral bounds.

9.6 Comparison of error requirements

Route	Required error control	What it retains
Individual means, Markov	$R+1$ embedding errors	Simplest version of the original triangle argument
Direct differences, Markov	R mean-difference errors	Improved constants without operator concentration
Geometric, exponential radius	Mean or difference concentration	Same deterministic theorem, better confidence dependence
Operator-aware	Mean-difference and second-moment errors	Population Pearson geometry, with additional estimation terms

A Variational derivation and exact empirical kernel form

For a reference P, the regularized functional is

$$\sup_{f\in\mathcal{H}_k}\left\{\mathbb{E}_Q f-\mathbb{E}_P f-\tfrac{1}{2}\mathbb{E}_P f^2-\tfrac{\lambda}{2}\|f\|^2\right\}.$$

Writing $A=T_P+\lambda I$ and $\Delta=\mu_Q-\mu_P$, complete the square:

$$\langle\Delta,f\rangle-\tfrac{1}{2}\langle f,Af\rangle=\tfrac{1}{2}\left\langle\Delta,A^{-1}\Delta\right\rangle-\tfrac{1}{2}\left\|f-A^{-1}\Delta\right\|_A^2.$$

The unique maximizer is $A^{-1}\Delta$ and the value is the score used throughout this note. Under suitable absolute continuity and square integrability, unrestricted optimization with no regularization yields $\frac{1}{2}\mathbb{E}_P[(dQ/dP-1)^2]$. RKHS restriction and regularization are part of the present score, so equality to unrestricted Pearson divergence is not presumed.

For a reference $X=(x_1,\dots,x_n)$ and query $Y=(y_1,\dots,y_q)$, let K_{XX},K_{XY},K_{YY} be the Gram blocks. With $\mathbf{1}$ of the appropriate dimension, put

$$B=\frac{\mathbf{1}_n^\top K_{XX}\mathbf{1}_n}{n^2}+\frac{\mathbf{1}_q^\top K_{YY}\mathbf{1}_q}{q^2}-\frac{2\mathbf{1}_n^\top K_{XY}\mathbf{1}_q}{nq},$$
$$h=K_{XY}\mathbf{1}_q/q-K_{XX}\mathbf{1}_n/n.$$

Then $B=\left\|\widehat{\Delta}\right\|^2$ and

$$\widehat{D}_\lambda=\frac{B-h^\top(K_{XX}+n\lambda I_n)^{-1}h}{2\lambda}. \tag{54}$$

To see this, define $\Phi a=\sum_a a_a\phi(x_a)$, so $\widehat{T}=\Phi\Phi^*/n$ and $\Phi^*\widehat{\Delta}=h$. Woodbury gives

$$(\widehat{T}+\lambda I)^{-1}=\lambda^{-1}\{I-\Phi(K_{XX}+n\lambda I_n)^{-1}\Phi^*\}.$$

Taking the quadratic form proves (54). This retains components outside the reference span through B. Omitting B, changing $n\lambda$ to λ, centering the operator, or dropping the half factor changes the score or its normalization. Small negative numerical roundoff is not a failure of the exact nonnegativity result.

B The optional bounded-differences calculation

For $F(X_1, \dots, X_n) = \|\widehat{\mu}_P - \mu_P\|$, Lemma 6.2 and Jensen give $\mathbb{E}F \le \sqrt{K/n}$. Replacing one observation changes F by at most $\|\phi(x) - \phi(x')\| / n \le 2\sqrt{K}/n$. Hence the sum of squared sensitivities is at most $4K/n$. McDiarmid's inequality gives

$$\mathbb{P}\{F - \mathbb{E}F \ge t\} \le \exp\{-nt^2/(2K)\}.$$

Taking $t = \sqrt{2K \log(1/a)/n}$ proves (40). If $k(x, y) \ge 0$ additionally, then $\|\phi(x) - \phi(y)\|^2 \le 2K$, so the same proof improves the radius to $\sqrt{K/n}(1 + \sqrt{\log(1/a)})$. This stronger constant uses pointwise nonnegativity, not merely positive semidefiniteness.

C A direct stability bound for the score

Under $\left\|\widehat{\Delta} - \Delta\right\| \le b$ and $\left\|\widehat{T} - T\right\|_{\mathrm{op}} \le e$, the resolvent identity gives

$$\widehat{A}^{-1} - A^{-1} = \widehat{A}^{-1}(A - \widehat{A})A^{-1}, \qquad \left\|\widehat{A}^{-1} - A^{-1}\right\|_{\mathrm{op}} \le e/\lambda^2.$$

Add and subtract $\left\langle \Delta, \widehat{A}^{-1}\Delta \right\rangle / 2$ in $\widehat{D}_\lambda - D$. If $z = \widehat{\Delta} - \Delta$, expanding the remaining quadratic form and using $\left\|\widehat{A}^{-1}\right\|_{\mathrm{op}} \le 1/\lambda$ gives

$$\boxed{|\widehat{D}_\lambda - D| \le \frac{2\,\|\Delta\|\, b + b^2}{2\lambda} + \frac{e\,\|\Delta\|^2}{2\lambda^2}.} \tag{55}$$

This bound makes mean and operator errors explicit but can be loose near the null if simplified further. At the exact null $\Delta = 0$, it reduces to the same $b^2/(2\lambda)$ upper bound. It is a complementary stability statement, not an assertion that $\widehat{D}_\lambda$ is unbiased for D.

D Normalization and finite-kernel score bounds

For $K = 1$, the principal common-individual-radius certificate is

$$\frac{2\varepsilon^2}{\lambda} \le \tau < \frac{(\delta - 2\varepsilon)^2}{2(1 + \lambda)}, \qquad \delta > 2\varepsilon\left(1 + \sqrt{\frac{1 + \lambda}{\lambda}}\right).$$

For Gaussian kernels, pointwise nonnegativity further bounds population and empirical squared embedding distances by $2K$, hence scores by K/λ. For a general bounded positive semidefinite kernel that can take negative values, the universal distance bound is instead $4K$ and the score bound $2K/\lambda$. Neither global cap asserts zero error for a realized finite same-regime comparison.